

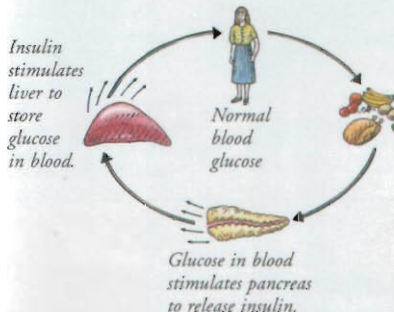
HORMONES AND THE ENDOCRINE SYSTEM



THE ENDOCRINE SYSTEM is one of the body's control systems. It consists of endocrine glands that produce chemicals called hormones, and release them into the bloodstream. The hormones act as chemical messengers and instruct specific areas of the body to carry out certain actions. Hormones usually work slowly and have a long-lasting action, regulating processes such as growth and reproduction.

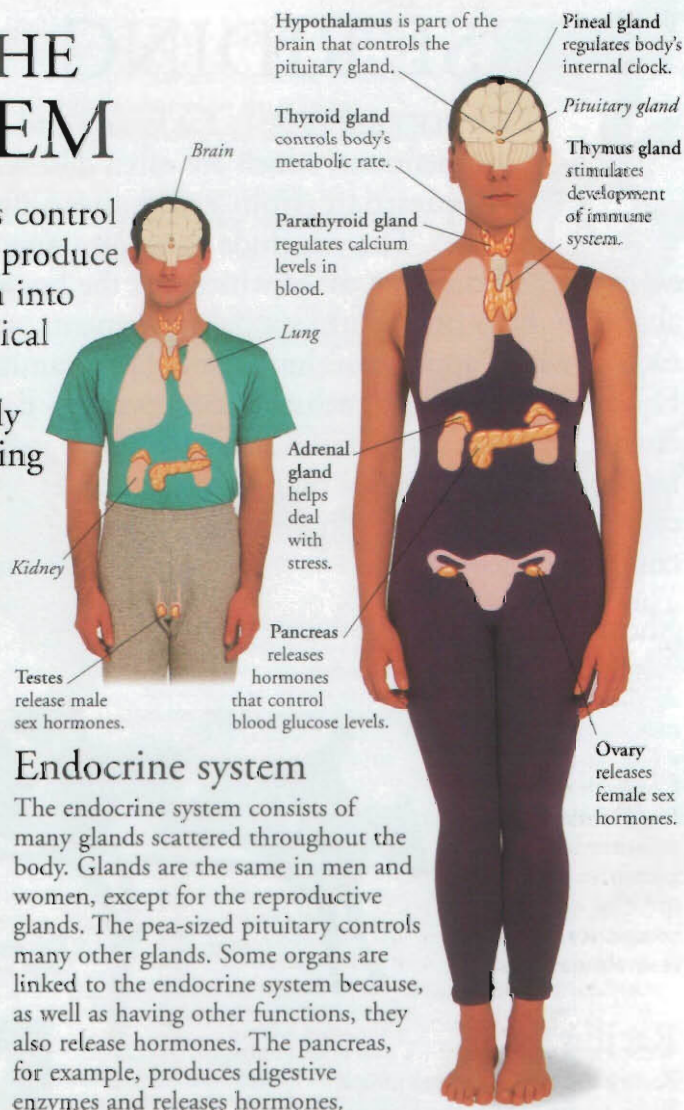
How a hormone works

The blood carries hormones throughout the body, but they only affect certain target cells within target tissues. The hormone attaches itself to a site on the surface of a target cell. This locking-on causes changes inside the target cells, which produce the required action. For example, the pancreas releases the hormone insulin in order to reduce levels of glucose molecules in the blood. Insulin does this by stimulating the body cells to take in glucose.



Hormone levels

Hormone levels in the blood are controlled by a feedback mechanism. For example, insulin is released from the pancreas in response to increased levels of glucose. Higher levels of insulin will then cause blood glucose levels to return back to normal. The lowered glucose levels "feed back" to the pancreas, which produces less insulin.



Endocrine system

The endocrine system consists of many glands scattered throughout the body. Glands are the same in men and women, except for the reproductive glands. The pea-sized pituitary controls many other glands. Some organs are linked to the endocrine system because, as well as having other functions, they also release hormones. The pancreas, for example, produces digestive enzymes and releases hormones.

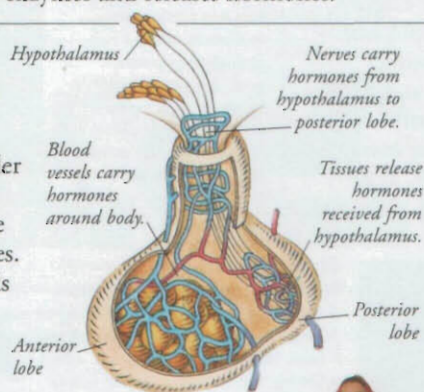


Diabetes

Diabetes is a condition that occurs when blood glucose levels become very high because the pancreas cannot produce enough insulin. Doctors can monitor blood glucose levels by measuring the amount of glucose in a urine sample.

Pituitary gland

The pituitary gland releases at least eight hormones. Some affect body functions directly, while the remainder stimulate other endocrine glands to produce hormones of their own. The pituitary gland has two parts, or lobes. The anterior lobe produces and sends hormones around the body; the posterior lobe releases hormones produced in the hypothalamus.



Jokichi Takamine

Japanese chemist Jokichi Takamine (1854-1922) was the first person to isolate a pure sample of a hormone.

Using extracts of adrenal glands, he prepared crystals of a substance that increased blood pressure in animals. This substance was later called adrenaline.



Prolactin

Prolactin is a hormone that is produced by the anterior lobe. It stimulates the production of milk when a woman breastfeeds her baby. When the baby sucks on the nipple, prolactin is immediately released from the mother's pituitary gland.



Growth hormone

The anterior lobe produces growth hormone, which encourages the body to grow. It works by stimulating the body's cells to divide. Although growth hormone affects all tissues, its main targets are bones and skeletal muscles. Growth hormone is most active in childhood and adolescence.



Adrenaline

If you have ever been frightened and felt your heart pounding, you have experienced the effects of adrenaline. It is a hormone that helps the body react to danger. When the adrenal glands release adrenaline, your breathing and heart rate speed up, and blood flows to your muscles so you can run from danger.

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